EduAssist AI — LLMOps Assignment 2

API Cost, Latency, and Prompt Version Management

Course: MLOps | Individual experiential learning assignment

Objective and design.

This assignment operationalizes an academic question-answering assistant by measuring request latency, token consumption, API cost, and prompt compliance. Three prompt versions are compared on the same five undergraduate machine-learning questions: v1 is a basic instruction, v2 adds an undergraduate-friendly role and style, and v3 requires a strict JSON object containing an answer, confidence label, and key points. The model used is Groq llama-3.1-8b-instant at USD 0.05 per million input tokens and USD 0.08 per million output tokens (rates as of 2 November 2026). The dataset is in simulated preview mode; all latency and token values are deterministically computed from the prompt and response lengths to demonstrate the LLMOps workflow without a live API key.

Task 1 — latency.

The notebook records question ID, prompt version, model, UTC start/end timestamps, latency, and response status for each of the 15 requests (3 versions × 5 questions). Latency statistics per version: v1 — mean 0.380s, median 0.365s, min 0.342s, max 0.421s, stdev 0.036s; v2 — mean 0.519s, median 0.534s, min 0.460s, max 0.568s, stdev 0.053s; v3 — mean 0.651s, median 0.672s, min 0.598s, max 0.693s, stdev 0.046s. The slowest request was v3/Q5 at 0.693s. v3 is consistently slower because the JSON schema instruction makes the prompt longer and the required structured output is longer to generate. In a live run, additional variance would come from network delay, model load, and output length.

Task 2 — token and cost economics.

The notebook extracts input and output token counts from each response and applies the formula: cost = (input_tokens / 1,000,000 × $0.05) + (output_tokens / 1,000,000 × $0.08). Results per version: v1 averages 23.2 input / 23.0 output tokens at $0.0000030/request ($0.030/month at 10,000 requests); v2 averages 39.2 / 23.0 tokens at $0.0000038/request ($0.038/month); v3 averages 59.2 / 33.2 tokens at $0.0000056/request ($0.056/month). v3 costs roughly 87% more per request than v1 due to the longer system prompt and JSON output. At this scale all versions are negligible in cost, but the ratio matters at higher volume. Cost optimizations: remove redundant prompt text, cap max_tokens, cache stable context, and route simple queries to a smaller model after validating answer quality.

Prompt comparison summary.

Prompt	Avg Latency (s)	Avg Input Tok	Avg Output Tok	Avg Cost/Req (USD)	Valid JSON	Monthly 10k (USD)
v1	0.380	23.2	23.0	$0.00000300	0%	$0.0300
v2	0.519	39.2	23.0	$0.00000380	0%	$0.0380
v3	0.651	59.2	33.2	$0.00000562	100%	$0.0562

Task 3 — prompt versioning and MLflow.

The experiment is named eduassist_llmops_assignment. For each prompt version the notebook logs avg_latency_seconds, avg_input_tokens, avg_output_tokens, avg_cost_per_request, and valid_json. The v3 validator checks that the response parses as JSON and that answer (str), confidence (high/medium/low), and key_points (list) are present — v3 achieves a 100% pass rate (5/5). v1 and v2 are plain-text prompts so valid_json is 0.0 by design. All three prompts are registered in MLflow's Prompt Registry under eduassist_academic_qa with commit messages: v1 'Register v1: baseline concise academic answer', v2 'Register v2: undergraduate-friendly structured instruction', v3 'Register v3: strict JSON schema output with confidence and key points'. The production alias is set to v3.

Recommendation and interpretation.

Prompt v3 is the recommended production candidate. It is the only version that enforces a machine-readable JSON contract, achieves 100% valid_json rate, and enables automated downstream parsing. The cost premium over v1 is $0.0000026/request — 87% more, but $0.026/month extra at 10,000 requests, which is negligible. The latency premium is 0.271s mean (71% slower than v1), which is acceptable for an academic Q&A; assistant but should be re-evaluated if SLA requirements tighten. v1 is the cheapest and fastest baseline; v2 improves readability for students but provides no structured output guarantee. For production, v3 should be promoted conditional on a live API run confirming the JSON pass rate and factual answer quality on a broader question set.

Limitations and conclusion.

All measurements in this submission use simulated preview values — latency is deterministically derived from prompt and response length rather than measured from a live API. JSON validity is verified structurally but does not assess factual correctness. A five-question sample is too small to establish robust quality or reliability. The experiment demonstrates the core LLMOps principle: prompts are production artifacts that must be versioned, evaluated against cost and latency budgets, and promoted through a traceable registry rather than edited informally.

References

Groq API documentation: https://console.groq.com/docs/openai

Groq model pricing: https://groq.com/pricing

MLflow documentation: https://mlflow.org/docs/latest/